

SW Engineering CSC 648 Fall 2025

Tutor Match

Team 02

Milestone 2

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1. Executive Summary

SFSU students often struggle to find trustworthy, course-specific academic help quickly. Generic tutor marketplaces don't reflect SFSU's curriculum, make it hard to verify that tutors are actually SFSU students, and bury students in irrelevant profiles. This wastes time, especially near midterms and finals, when targeted help matters most.

TutorConnect is a desktop-first web application exclusively for the SFSU community. Students can **search by SFSU course code** (e.g., "CSC 340"), browse **approved** tutor listings, view profiles with clear availability, and initiate contact through a **single, in-site message**, so no email or chat is required. Student tutors create or update listings that remain **pending** until an **admin** approves them for quality and policy compliance. No payment features are implemented or simulated in the UI. The platform is **SFSU-only** for posting and messaging (accounts must use a valid @sfsu.edu). Search pivots around **course codes** and subjects students actually take at SFSU, helping them match quickly and confidently. Admin approval plus a simple flagging mechanism improves safety and quality.

Team 02 is a cross-functional group of SFSU student software engineers (front-end, back-end, and DevOps). We're building a maintainable, standards-based system (React (Vite) + Node + MySQL + Python) that easily evolves in later milestones with instructor feedback.

2. Main Data Items and Entities

User (abstract)

Meaning: Any signed-in person; specialized into Student, Tutor, Admin.

Items: userId (M), fullName (M), email (M; must end with @sfsu.edu for posting/messaging), role (M: STUDENT/TUTOR/ADMIN), status (M: active/suspended), createdAt (M)

Notes: Email rule enforces SFSU-only posting & messaging.

Visibility: Public minimal (name on listings), full to Admin.

Registered User / Student (specialization of User)

Items: studentId (M; = userId), major (O), graduationTerm (O), profilePhotoId (O)

Used for: Dashboard views, (future) reviews.

Tutor (specialization of User)

Items: tutorId (M; = userId), verificationStatus (M: pending/approved/rejected), hourlyRate (O; display only), profilePhotoId (O)

Used for: Listings, availability, moderation.

Course (SFSU-specific)

Meaning: Exact SFSU course metadata to power course-code search.

Items: courseId (M), department (M; e.g., CSC), courseNumber (M; e.g., 340), classNumber (O; SFSU Class #), term (O; e.g., Fall 2025), instructorName (O), title (O)

Notes: Used in search filters and tutor listings.

Subject

Meaning: Topic taxonomy (e.g., Data Structures, Writing).

Items: subjectId (M), name (M), category (O), description (O)

Relations: Many-to-many with Tutor; optional mapping to Course.

Listing (Tutor's public card)

Meaning: Public searchable card created by a Tutor; moderated. This is the tutor's way of showing which courses they will be able to tutor with

Items:

listingId (M), tutorId (M),
title (M), bio (M; short description),
subjects[] (M: ≥ 1), courses[] (O but recommended for SFSU specificity),
modality (M: inPerson/online/hybrid),
status (M: draft/pending/approved/rejected),
media[] (O; images/video refs),
createdAt (M), updatedAt (M)

Notes: On create/edit → Pending until Admin approval.

Availability

Meaning: Time windows a Tutor offers (simple recurring blocks in M2). This IS NOT scheduling of sessions, but rather a way to show the user what days the tutor is available before they select the tutor.

Items:

availabilityId (M), tutorId (M),
daysOfWeek (M; e.g., MON/WED/FRI), timeWindow (M; e.g., 18:00–21:00),
slotLengthMinutes (M; e.g., 60),
modeOfInstruction (M: online/inPerson/hybrid),
effectiveStartDate (O), effectiveEndDate (O),
locationNote (O), meetingLinkTemplate (O)

Notes: Keep simple in M2; advanced recurrence can come later.

Message (one-round tutee → tutor)

Meaning: Single in-site message to initiate contact; no chat.

Items:

messageId (M), fromUserId (M; student), toUserId (M; tutor),
listingId (M), body (M; short inquiry), createdAt (M),
status (M: sent/read/archived)

Rules: One-round rule — block additional messages from same student to same tutor.

Review (P2, later milestone)

Items: reviewId (M), tutorId (M), studentId (M), rating (M; 1–5), comment (O), visibilityStatus (M: public/hidden/flagged), createdAt (M)

Moderation: Subject to Admin hide/flag.

Report/Flag

Meaning: Community moderation signal.

Items: reportId (M), reporterId (M), targetType (M: listing/media/review/user), targetId (M), reason (M), actionTaken (O), createdAt (M)

Bookmark (P2)

Meaning: Private “save for later.”

Items: bookmarkId (M), userId (M), targetType (M: listing/tutor), targetId (M), createdAt (M), note (O)

3. Functional Requirements (Prioritized)

Priority 1 (must-have)

Unregistered users (public)

1. (M1 5.1.1) Public browse
 - 5.1.1.1 Show a paginated list of Approved listings only.
 - 5.1.1.2 Each card shows: title, tutor name, subjects, short bio snippet, modality.
2. (M1 5.1.2) Public profile view
 - 5.1.2.1 Profile displays: bio, subjects[], courses[] (SFSU codes), availability snapshot, approved media.
 - 5.1.2.2 No contact info beyond the in-site Message entry point.
3. (M1 5.1.4) SFSU course-code search (SFSU-specific)
 - 5.1.4.1 Search by department + courseNumber (e.g., “CSC 340”).
 - 5.1.4.2 Results must only include listings that explicitly include that course in courses[].
4. (M1 5.1.5) Filter and sort
 - 5.1.5.1 Filters: subject, modality, simple availability (e.g., “Evenings”).
 - 5.1.5.2 Sort: relevance (default), most recent.

Registered users (SFSU email required)

5. (M1 5.2.2) SFSU registration rule
 - 5.2.2.1 Only @sfsu.edu accounts can post listings or send messages.
 - 5.2.2.2 Public browse remains open to all.
6. (M1 5.2.3) Sign-in / sign-out / session
 - 5.2.3.1 Maintain authenticated session with secure cookie.
 - 5.2.3.2 Show a basic user menu (Profile, Dashboard, Sign-out).

7. (M1 5.2.5) One-round in-site message (tutee → tutor)
 - 5.2.5.1 Allow exactly one message per student→tutor pair.
 - 5.2.5.2 Block duplicates with a clear UI notice (“You already contacted this tutor.”).
 - 5.2.5.3 Store status (sent/read).
8. (M1 5.2.8/5.2.9) Create/edit listing → Pending
 - 5.2.8.1 Tutor can create/edit title, bio, subjects[], courses[], modality, media[].
 - 5.2.9.1 On submit/edit, set status = pending (hide from public) until Admin approval.
9. (M1 5.2.11) Availability management (simple in M2)
 - 5.2.11.1 Tutors can add simple recurring windows (days).
 - 5.2.11.2 Availability snapshot appears on the profile.
10. (M1 5.1.3) Keyword/subject search
 - 5.1.3.1 Free-text search across title + bio (SQL %LIKE%).
 - 5.1.3.2 Subject facet narrows results to subjects[].

Admin / Moderators

11. (M1 5.3.1) Approve/reject listings
 - 5.3.1.1 Queue of pending listings with preview of text/media.
 - 5.3.1.2 Action sets status = approved | rejected (+ optional note).
12. (M1 5.3.3) Manage and resolve flags
 - 5.3.3.1 View reports with reason; set actionTaken and mark Resolved.

Priority 2 (should-have)

Registered users

1. (M1 5.2.1) Inherits public functions
 - 5.2.1.1 Signed-in users get the same browse/profile/search plus saves and dashboards.
2. (M1 5.2.4) Profile management
 - 5.2.4.1 Edit name, photo, and (if Tutor) set Tutor role flag.
3. (M1 5.2.6) Tutee dashboard
 - 5.2.6.1 Show sent message(s) with status; show saved listings.
4. (M1 5.2.7) Save/unsave tutor
 - 5.2.7.1 Toggle bookmark on listings; maintain private shortlist.
5. (M1 5.2.10) Tutor dashboard
 - 5.2.10.1 Show listing status (Draft/Pending/Approved/Rejected) and view counters.
6. (M1 5.2.12) Report/flag content
 - 5.2.12.1 Allow a signed-in user to flag a listing/media with a short reason.

Admin / Moderators

7. (M1 5.3.2) Moderate content and users
 - 5.3.2.1 Hide/remove listings or media; suspend/reinstate accounts (with audit note).

8. (M1 5.3.4) Admin search & audit trail
 - 5.3.4.1 Search by user/listing; show who/what/when for moderation actions.

Priority 3 (opportunistic / later)

Registered users

1. (M1 5.2.13) Simple rating/review (post-help)
 - 5.2.13.1 Student can leave rating + short comment on a tutor; subject to moderation.

Will add more P3 feature as we come up with them

4. UI Storyboards

This section presents black-and-white UI storyboards for the main use cases of the SFSU Tutor App. Each storyboard illustrates the user flow for a specific use case, focusing on layout, navigation, and main functions (P1 priority). The mockups were designed in Figma for the browser version and mobile browser version as well (no mobile app version).

4.1 Notes on Admin UI

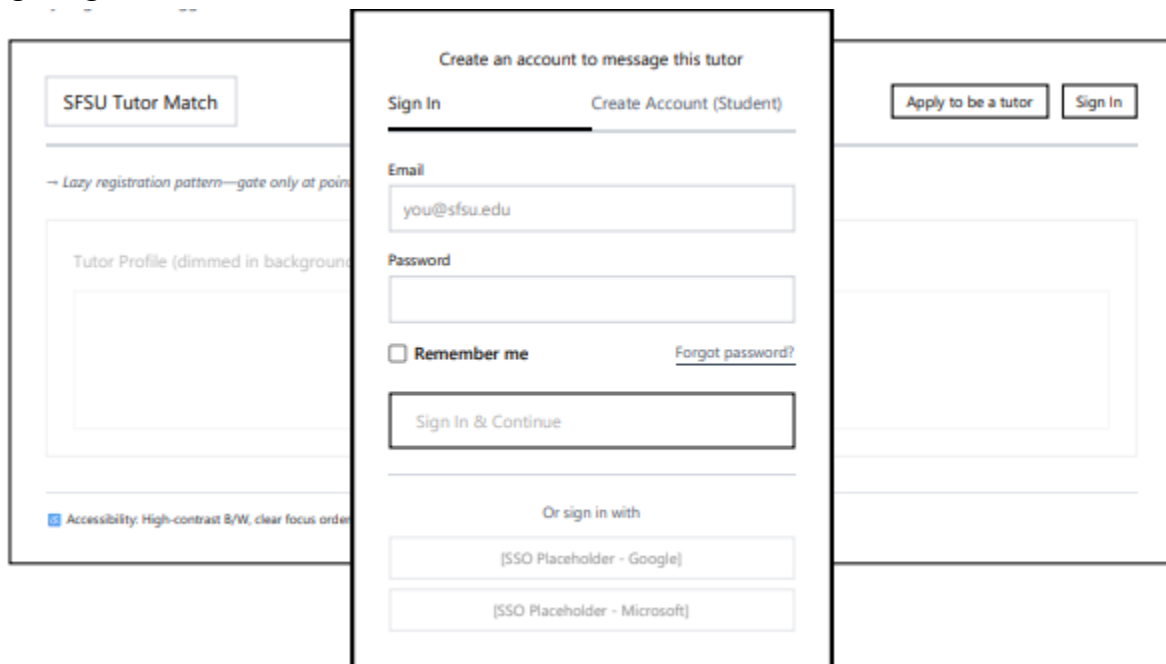
The admin interface is not included in this section, since administrative tasks will be performed using MySQL Workbench..

4.2 Storyboards for Main Use Cases

4.2.1 Login/Registration, Apply as Tutor

Related Use Cases: Precondition for all UC flows

Fig.1 Login



Profile content dimmed...

Create an account to message this tutor

Sign In

Create Account

Email

you@sfsu.edu

Password

☐ Remember me

[Forgot?](#)

Sign In & Continue

Or sign in with

[Google SSO]

[Microsoft SSO]

→ Modal centered on mobile viewport

→ Tabs switch between Sign In & Create Account

Fig.2 Apply as Tutor

SFSU Tutor Match

Find TutorsSubjectsHow it Works

Apply to be a tutorSign In

—Single-column for scanability; required fields clearly marked with asterisks

1

2

Step 1 of 2: Basic Information

Become a Tutor

Basic Info * Required

First Name *

Last Name *

Email *

you@sfsu.edu

Phone

(555) 123-4567

Optional — for contact purposes only

Profile Photo *

Drag and drop your photo here

or

Upload

Accepted: JPG, PNG (max 10MB)

Resume/CV *

Drag and drop your resume here

or

Upload

Accepted: PDF only (max 10MB)

Bio *

Tell students about your teaching experience, your approach, and what makes you a great tutor.

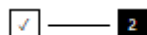
Min: 100 characters, max: 600 characters

Save Draft

Continue

Accessibility: High contrast B/W, clear focus order, large tap targets. Required fields clearly marked with asterisks.

→ Meets 'days only' requirement; no time pickers.



Step 2 of 2: Courses & Availability

Courses & Teaching Details

Courses Taught *

Add the SFSU courses you can tutor

Subject	Course Code
Select subject ▼	e.g., CSC 340
+ Add Course	

CSC 340 - Software Engineering ✕

CSC 220 - Data Structures ✕

MATH 226 - Calculus I ✕

Mode of Instruction *

How do you prefer to tutor?

☐ In-person	Meet students on campus or at the library
☒ Online	Conduct sessions via Zoom or video call
☐ Both	Flexible — either in-person or online

Availability *

Times handled later; days only for now.

Mon Monday	Tue Tuesday	Wed Wednesday	Thu Thursday
Fri Friday	Sat Saturday	Sun Sunday	

[Back](#)

[Submit Application](#)

— Clear status; no scheduling features.



Thanks! Your application is under review.

We've received your tutor application and will review it within 2-3 business days. You'll receive an email once your profile is approved.

What's Next?

- 1 Our team will review your resume, bio, and course qualifications
- 2 We may reach out for additional verification or information
- 3 Once approved, your profile will be live and students can contact you
- 4 You'll receive an email notification when students message you

[View Application](#)[Return to Home](#)

Questions? Contact us at tutors@sfsu.edu

1

2

Become a Tutor

Step 1 of 2: Basic Info

Basic Info * Required

First Name *

Last Name *

Email *

you@sfsu.edu

Phone

(555) 123-4567

Optional

Profile Photo *

Tap to upload

Upload

JPG, PNG (max 5MB)

Save Draft

Continue

→ Single-column layout for mobile scannability

→ Upload areas optimized for touch

Become a Tutor

Step 2 of 2: Courses & Availability

Courses Taught *

Add SFSU courses you can tutor

Subject

Select subject



Course Code

e.g., CSC 340

+ Add Course

CSC 340 - Software Engineering



CSC 220 - Data Structures



MATH 226 - Calculus I



Mode of Instruction *

How do you prefer to tutor?



In-person

Meet on campus or at library



Online

Via Zoom or video call

Back

Submit Application

→ Course chips stack vertically on mobile

→ Days-only 2-column grid

Application Submitted



Thanks! Your application is under review.

We've received your tutor application and will review it within 2-3 business days.

What's Next?

- 1 We'll review your resume, bio, and qualifications
- 2 We may reach out for additional verification
- 3 Once approved, your profile will be live
- 4 You'll get email notifications when students message you

[Return to Home](#)

[View Application](#)

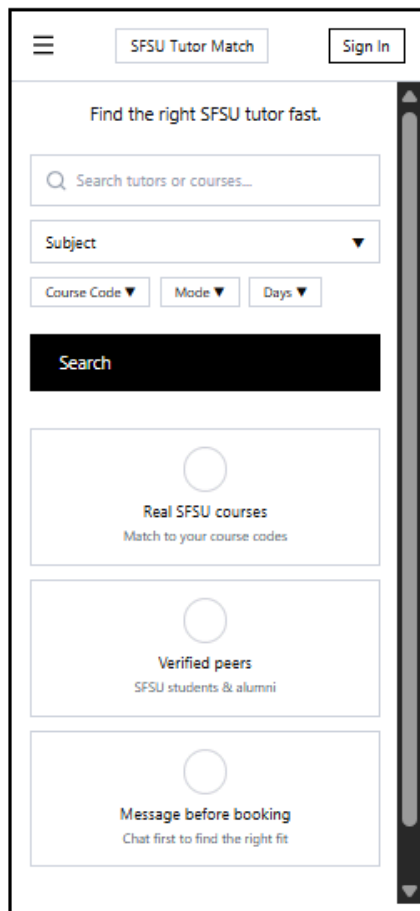
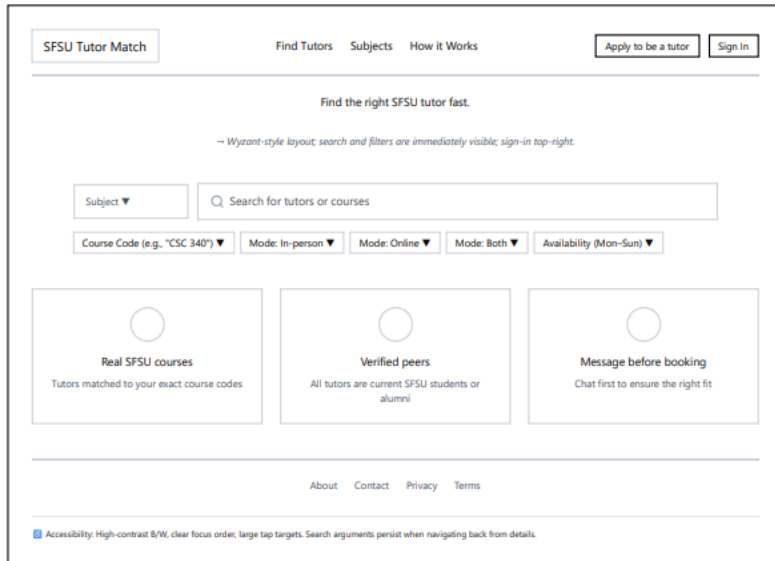
Questions? tutors@sfsu.edu

→ Clear success state

→ Numbered steps for clarity

4.2.2 UC-1 — Search by SFSU Course Code, Browse, Filter, View Tutor Profile, and Message Tutor

Fig.3 Landing Page



→ Hamburger for nav menu

→ Stacked filters for mobile

Fig.4 Search Results Page

SFSU Tutor Match

Find TutorsSubjectsHow it Works

Apply to be a tutorSign In

→ Filters persist; availability shows days only; college-friendly list.

Showing 1–12 of 128 tutorsSort: Relevance ▼

Filters

Subject

Computer Science

Mathematics

Biology

Course Code

e.g., CSC 340

Mode of Instruction

In-person

Online

Both

Availability

Days only

Mon

Tue

Wed

Thu

Fri

Sat

Sun

Tutor Name 1

☆☆☆☆☆

Computer ScienceCSC 340Online

Short bio about tutor experience and teaching style...

View ProfileContact Tutor

Tutor Name 2

☆☆☆☆☆

Computer ScienceCSC 340Online

Short bio about tutor experience and teaching style...

View ProfileContact Tutor

Tutor Name 3

☆☆☆☆☆

Computer ScienceCSC 340Online

Short bio about tutor experience and teaching style...

View ProfileContact Tutor

Tutor Name 4

☆☆☆☆☆

Computer ScienceCSC 340Online

Short bio about tutor experience and teaching style...

View ProfileContact Tutor

Accessibility: High-contrast B/W, clear focus order, large tap targets. Keep search arguments persistent.

128 tutors found

Sort: Relevance

Tutor Name 1

☆☆☆☆

CS

CSC 340

Online

Senior CS major with 3 years experience...

View

Contact

Tutor Name 2

☆☆☆☆

CS

CSC 340

Online

Senior CS major with 3 years experience...

View

Contact

Tutor Name 3

☆☆☆☆

CS

CSC 340

Online

Senior CS major with 3 years experience...

View

Contact

Tutor Name 4

☆☆☆☆

CS

CSC 340

Online

Senior CS major with 3 years experience...

Filters

Subject

☐ Computer Science

☐ Mathematics

☐ Biology

Course Code

e.g., CSC 340

Mode of Instruction

☐ In-person

☐ Online

☐ Both

Apply Filters

→ Bottom sheet with all filters

→ Days only, no time pickers

Fig.5 One-way message to tutor

SFSU Tutor Match

Find TutorsSubjectsHow it Works

Apply to be a tutorSign In

→ M2 scope = messaging initiation only (no session scheduling).

Message Tutor

Alex Johnson

Computer Science Major, Senior

☆☆☆☆

CSC 340CSC 220Online

New Conversation

Start a conversation below

Introduce yourself and the course (e.g., CSC 340)

Example: Hi Alex! I'm taking CSC 340 this semester and struggling with design patterns. Would you be available to help?

🔔 We'll email you when they reply

Send

Tips for your first message

- Mention the specific course you need help with
- Briefly describe what you're struggling with
- Ask about their availability
- Be polite and professional

 Accessibility: High-contrast B/W, clear focus order, large tap targets. Textarea clearly labeled.

←

Message Tutor

Alex Johnson

CS Major, Senior

☆☆☆☆

CSC 340

CSC 220

Online

Start a conversation below

Introduce yourself and the course (e.g., CSC 340)

Example: Hi! I'm taking CSC 340 this semester and need help with design patterns...

💡 We'll email you when they reply

Send

→ Tutor info always visible

→ Placeholder suggests course mention

4.2.3 UC-5 Save Tutors & Use the Tutee Dashboard

SFSU Tutor Match

[Find Tutors](#)[Subjects](#)[How It Works](#)

[Apply to be a tutor](#)[Sign in](#)

— Dashboard shows who the tutor saved and who they reached out to; contacting triggers lay registration if not signed in; one-round message enforced

Welcome, Sarah Chen
Manage your saved tutors and messages

[Saved](#) 3 [Messages](#) 2

Saved Tutors

Alex Johnson

[CSC 340](#)[CSC 220](#)[Online](#)[In-person](#)

[View Profile](#)[Contact Tutor](#)

[Remove](#)

Maria Rodriguez

[MATH 226](#)[MATH 227](#)[Online](#)

[View Profile](#)[Contact Tutor](#)

[Remove](#)

David Kim

[BIOL 230](#)[CHEM 115](#)[In-person](#)

[View Profile](#)[Contact Tutor](#)

[Remove](#)

No saved tutors yet
Browse tutors and bookmark the ones you're interested in
[Find Tutors](#)

Messages (when Messages tab is active)

Tutor	Course	Sent Date	Status	Action
Alex Johnson	CSC 340	Nov 1, 2024	Resend	Open
Maria Rodriguez	MATH 226	Oct 26, 2024	Send	Open

No messages yet
[Find Tutors](#)

[Accessibility](#) [Clear site navigation](#) [Improve focus support](#) [Site Maps](#) [Feedback](#) [Description of icons](#)

— Defcon one-round rule; no live chat thread

[← Back to Messages](#)

Message Detail



Alex Johnson
Computer Science Major, Senior

[CSC 340](#) [CSC 220](#) [Online](#)

 [Save](#)

 **You already contacted this tutor.** You can't send another message through the platform. The tutor may reach out to you directly.

Your Message [Read](#)

Sent on Nov 1, 2024 at 2:30 PM

Hi Alex!

I'm taking CSC 340 this semester and I'm struggling with understanding design patterns, especially the Observer and Factory patterns. I saw that you tutor this course and have great reviews.

Would you be available to help? I'm flexible with online sessions and prefer evenings during the week. Let me know your availability and rates.

Thanks!

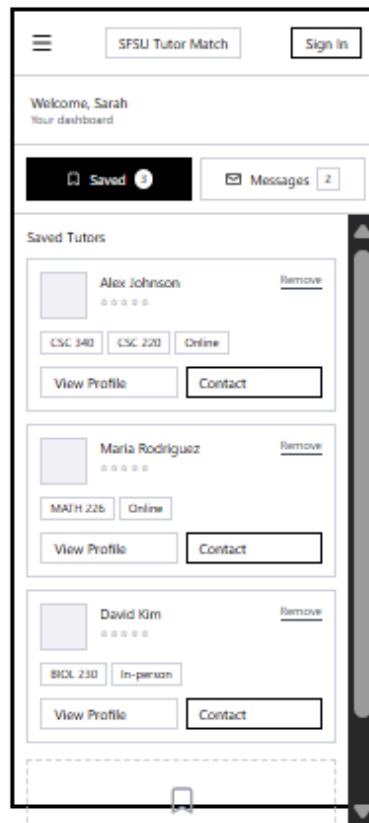
[View Profile](#) [Back to Messages](#)

What happens next?

- The tutor can see your message and may respond via email or phone
- Check your SFSU email for replies
- If you don't hear back within 3 days, try another tutor
- Remember to keep all communication appropriate and respectful

Mobile: Tutee Dashboard Overview (375x812)

→ Mobile friendly; same labels/structure; no time pickers, no video; stacked cards with full width buttons.



→ Segmented tabs for mobile

→ Stacked cards, full-width actions

Accessibility: Large tap targets (min 44x44px), clear labels, scrollable content area.

4.2.4 UC-8 — Monitor Listing & Inquiries in Tutor Dashboard

SFSU Tutor Match

Find TutorsSubjectsHow it Works

Apply to be a tutorSign In

→ Tutor sees all Messages and current Listing status; no sessions/payments.

Welcome, Alex Johnson

Manage your tutoring profile and incoming messages

 Inbox

5 new messages

Student	Course	Received	Status	Action
● Sarah Chen	CSC 340	Nov 1, 2024	New	Open
● Michael Torres	CSC 220	Oct 31, 2024	New	Open
Emma Wilson	CSC 340	Oct 30, 2024	Read	Open
James Lee	MATH 226	Oct 28, 2024	Read	Open
Sofia Martinez	CSC 220	Oct 27, 2024	Read	Open

 Listing Status

Current Status: **Approved**

Your profile is live and visible to students

👤 128 profile views this month

View Listing

Edit Listing

Listing Status Options

Draft

Pending

Approved

Rejected

Not submitted

Under review

Live

Needs changes

 Accessibility: Clear table structure, status indicators with text labels, keyboard navigation support.

→ No in-app chat; message receipt only; tutor can mark read.

[← Back to Inbox](#)

Message from Student



Sarah Chen
SFSU Student

Interested in: CSC 340

Message

Received on Nov 1, 2024 at 2:30 PM

Now

Hi Alex!

I'm taking CSC 340 this semester and I'm struggling with understanding design patterns, especially the Observer and Factory patterns. I saw that you tutor this course and have great reviews.

Would you be available to help? I'm flexible with online sessions and prefer evenings during the week. Let me know your availability and rates.

Thanks!

Actions

Mark as Read

[Back to Inbox](#)

Suggested Next Steps

This platform does not support in-app messaging or scheduling.

If you're interested in working with this student, you may respond externally using appropriate communication channels (e.g., your SFSU email, phone).

- Keep all communication professional and appropriate
- Arrange meeting times and locations directly with the student
- Discuss rates and payment methods outside the platform
- Follow SFSU guidelines for peer tutoring

Note: SFSU Tutor Match is a connection platform only. We do not facilitate payments, scheduling, or session management.

SFSU Tutor Match

Sign in

Welcome, Alex

Tutor Dashboard

Listing Status

Status: **Approved**

Profile is live

128 views this month

View Listing

Edit Listing

Inbox

5 new

• Sarah Chen

CSC 340

Nov 1

Open Message

• Michael Torres

CSC 220

Oct 31

Open Message

Emma Wilson

CSC 340

Oct 30

Read

— Listing status at top

— Message cards with clear status

5. High-level Architecture

5.1 Overall architecture (MVC + deployment)

Client (View): Vite/React SPA (desktop-first; mobile friendly). Static assets served by Nginx.

Server (Controller + Model): FastAPI (Python) app exposing REST endpoints; controllers handle requests/validation; services run domain logic; data layer uses a DB access library/ORM.

Database: MySQL (prod + test). SQL schema mirrors Section 2 logical items.

Reverse proxy: Nginx terminates TLS, serves the SPA, and proxies /api/* to FastAPI (Uvicorn/Gunicorn).

Secrets & config: environment variables on server; never committed to Git.

Separation of concerns: UI (React) has no SQL; all data access goes through API; DB changes are versioned via migrations.

Flow of our application:

Browser (React SPA)

↕ HTTPS

Nginx (TLS, static, reverse proxy)

↕ HTTP (internal)

FastAPI (Controllers → Services → Repos)

↕ SQL

MySQL (prod / test)

5.2 Database organization

*Names align with Section 2. Types/constraints omitted here on purpose; this is a high-level list.

Users:

user_id, full_name, email (SFSU enforced for posting/messaging), role (STUDENT/TUTOR/ADMIN), status (active/suspended), created_at

Tutors (one-to-one with Users where role = TUTOR):

tutor_id (= user_id), verification_status (pending/approved/rejected), hourly_rate (display only), profile_photo_media_id (optional)

Students (one-to-one with Users where role = STUDENT):

student_id (= user_id), major (optional), graduation_term (optional), profile_photo_media_id (optional)

Subjects:

subject_id, name, category (optional), description (optional)

Courses (SFSU):

course_id, department (e.g., CSC), course_number (e.g., 340), class_number (optional), term (optional), instructor_name (optional), title (optional)

Listings (public tutor cards; moderated):

listing_id, tutor_id, title, bio, modality (in_person/online/both), status (draft/pending/approved/rejected), created_at, updated_at

ListingSubjects (many-to-many link):

listing_id, subject_id

ListingCourses (many-to-many link; enables course-code search):

listing_id, course_id

Availability (days only for M2):

availability_id, tutor_id, days_of_week (e.g., ["Mon", "Wed", "Fri"]), mode_of_instruction (online/in_person/hybrid), note (optional)

Messages (one-round tutee → tutor):

message_id, from_user_id (student), to_user_id (tutor), listing_id, body, status (sent/read), created_at

Reviews (later milestone; optional now):

review_id, tutor_id, student_id, rating, comment (optional), visibility_status (public/hidden/flagged), created_at

Flags (reports):

flag_id, reporter_id, target_type (listing/media/review/user), target_id, reason, action_taken (optional), created_at

Bookmarks (Saved for later):

bookmark_id, user_id, target_type (listing/tutor), target_id, created_at, note (optional)

MediaAssets (metadata for uploads):

media_id, owner_user_id, kind (profile_photo/listing_media/resume), storage_path, original_filename, mime_type, size_bytes, created_at

Indexing (initial): listings(status), listing_courses(course_id), subjects(name), courses(department, course_number), messages(to_user_id, status); consider compound indexes for search filters as usage data arrives.

5.3 Media storage

Decision: For M2 we store files in the server file system (under a dedicated uploads directory) and metadata in MediaAssets. This allows for simpler backups, faster streaming, and avoids bloating DB.

Supported types now: images (jpg/png/webp) and PDF (resume/CV).

Limits: file size cap (e.g., 5–10 MB), server-side validation of MIME and extension.

Future option (later milestone): migrate to S3 with the same MediaAssets metadata contract (drop-in change).

5.4 Search & filter architecture (implementation summary)

Approach: SQL queries with LIKE/%LIKE% over selected text fields; no external search engine in M2. All queries use prepared statements (param binding) to prevent SQL injection.

User-facing inputs (from Search UI):

Subject (facet) → subjects.name via ListingSubjects join

Course Code (e.g., “CSC 340”) → exact/normalized match on courses.department + courses.course_number via ListingCourses

Mode of Instruction → listings.modality

Keyword (free text) → %LIKE% over listings.title + listings.bio (concatenated)

Availability (days only) → availability.days_of_week contains any of the selected days

DB terms searched with %LIKE%: listings.title, listings.bio (optionally subjects.name, courses.title in a second pass if needed).

Organization: Controllers validate & normalize inputs (e.g., trim, uppercase dept codes), Services build query specs, Repos compose SQL with safe parameters.

Sorting: Default Relevance (see 5.5) then Most Recent (listings.updated_at DESC).

Pagination: request params page, page_size with caps; return total_count for UX.

Result contract: stable JSON array of Listing cards with tutor summary and chips (Subject/Course/Mode), matching the storyboarded components.

5.5 Non-trivial logic (simple relevance & rules)

Relevance scoring (lightweight, SQL-friendly):

Exact Course Code match (department + course_number) → highest weight

Subject name match → medium weight

Keyword in title → medium weight

Keyword in bio → low weight

Tie-breaker: recently updated listings first

Implementation: compute a score via CASE-based weights in SQL or a service-level post-score; order by score DESC, updated_at DESC.

One-round messaging enforcement: API checks if a from_user_id → to_user_id pair already exists; if yes, block new sends with a clear message.

SFSU-only enforcement for posting/messaging: server verifies email domain before allowing create listing or send message actions.

5.6 Tools & frameworks (no new tools without CTO approval)

Frontend: Vite + React (no change).

Backend: FastAPI (Python), Uvicorn/Gunicorn, SQLAlchemy or equivalent repo pattern (no change).

DB: MySQL (prod + test) (no change).

Proxy/Hosting: Nginx on AWS EC2 (no change).

New tools/frameworks: None added in M2. Any new addition would be documented here and pre-approved by the CTO in writing before use.

5.7 Notes & constraints

Out of scope for M2: live sessions, time-of-day scheduling, payments, full chat. Availability is days only.

Consistency: API uses clear camelCase field names aligned with Section 2; DB tables use snake_case; mapping is documented in code.

Security basics: input validation, size/type checks on uploads, least-privilege DB credentials, and request logging for moderation trails.

6. Key Risks

Skills risks: Back end and front end teams need to delve deeper into how to use AI tools properly

Schedule risks: No schedule risks, Team 02 gets the job done on time (going forward)

Technical risks: No technical risks at this time

Teamwork risk: Making sure everyone understands what is being asked of us in each milestone, and assuring the highest quality of work

Legal/content risks: None

7. Project Management

Approach & Roles: We managed Milestone 2 with a lightweight, delivery-focused process. Each high-level requirement from M1 is owned by one role: Backend Lead (API/DB and server deploy), Frontend Lead (VP search page and mockups/storyboards), Scrum Master (QA and meeting management), and Team Lead (PM + DevOps). The Team Lead also coordinates handoffs and final QA.

Planning cadence:

- Weekly planning (30 min): set this week's cards and due dates; update priorities (P0/P1/P2).
- Daily check-in (10 min, async on Trello): each owner posts status: Done / Doing / Blocked.
- Mid-week integration check (15 min): FE↔BE verify the API still matches the contract.
Pre-submit review (30–45 min): run the VP checklist on the live URL and finalize the doc.

Task tracking (Trello).

We use one Trello board with lists Milestone 2 (templates), Today, This Week, Later. Cards exist per role: Backend — API+DB; Frontend — VP Page+Storyboards; Member A — Data+QA; Member B — PM/Docs+DevOps; Integration & Freeze; Submission (VP Email). Each card has a checklist, an owner, labels (P0/P1/P2), and a due date. Attachments include links to PRs, live URLs, and docs. Calendar view is used to see all due dates at a glance.

Quality & reviews (to be implemented):

Every PR requires one reviewer. Member A owns a VP QA checklist (category filter, text %LIKE, combined search, empty state, inputs persist, live URL opens). We run the checklist on test then prod. Member B ensures the Nginx/proxy note, credentials note (no secrets in Git), and the VP submission email are complete.

Communication:

Discord: day-to-day, blockers, and quick decisions, and voice call for meetings

Trello comments: task-specific discussion and links.

Escalation: blockers >24h go to Team Lead; if still blocked, scope is trimmed (P2 → later) to protect the VP and submission.

8. Use of GenAI Tools

Plan to use Figma AI Make to turn wireframes into HTML/CSS code

Used GrammarlyAI to proofread grammar in this document

9. Team Lead Checklist

1. So far all team members are fully engaged and attending team sessions when required: ON TRACK
2. Team ready and able to use the chosen back and front end frameworks and those who need to learn are working on learning and practicing: ON TRACK
3. Team reviewed suggested resources before drafting Milestone 2 : DONE
4. Team lead checked Milestone 2 document for quality, completeness, formatting and compliance with instructions before the submission: DONE
5. Team lead ensured that all team members read the final Milestone 2 document and agree/understand it before submission: DONE
6. Team shared and discussed experience with genAI tools among themselves: DONE

10. Team Members and Roles

Team Lead: [Ranjiv Jithendran](#)

Front-End Lead: [Adea Mulaku](#)

Back-End Lead: [Bao Than](#)

GitHub Master: [Dhvanil Bhagat](#)

Scrum Master: [Harry Wong](#)

11. Team Lead Checklist

1. So far all team members are fully engaged and attending team sessions when required: **ON TRACK**
2. Team found a time slot to meet outside of the class: **DONE/OK**
3. Team ready and able to use the chosen back and front end frameworks and those who need to learn are working on learning and practicing: **ON TRACK**
4. Team reviewed class slides on requirements and use cases before drafting Milestone 1: **DONE/OK**
5. Team reviewed non-functional requirements from “How to start...” document and developed Milestone 1 consistently: **DONE/OK**
6. Team lead checked Milestone 1 document for quality, completeness, formatting and compliance with instructions before the submission: **DONE/OK**
7. Team lead ensured that all team members read the final M1 and agree/understand it before submission: **DONE/OK**
8. Team shared and discussed experience with GenAI tools among themselves: **DONE/OK**
9. Github organized as discussed in class (e.g. master branch, development branch, folder for milestone documents etc.): **DONE/OK**